

I am a passionate engineering student expected to graduate from KU Leuven in June 2026. I am a multi-faceted student with a strong incline towards AI architecture, Logic design and Software-Hardware co-design. My background provides me a deep understanding of current trends and designs of Semiconductors circuitry. I am now looking to find new opportunities in Belgium and abroad.

EDUCATION

M. Sc. in Electrical Engineering, chip design and electronics September 2024 - June 2026
KU Leuven, Magna cum Laude Leuven, Belgium

B. Sc. in Engineering, Electricity and Computer Science September 2021 - June 2024
UCLouvain, cum Laude Louvain-la-Neuve, Belgium

SKILLS

Hardware/EDA	Cadence, Altium, KiCad, Verilog, SystemVerilog, Chisel, Vivado, Quartus, Fusion 360
Software/AI	Python, PyTorch, LLM, HuggingFace, C, ASM, JAVA, Matlab, Simulink, Git, Bash
Proven Skills	Embedded Systems, FPGA, Reverse Engineering, Hardware Security, EMFI, AI Accelerators
Languages	French (Mother Tongue), English (C1), Dutch (B2), German (A1)

TECHNICAL EXPERIENCE

VULNERABILITY ANALYST INTERN June 2025 - August 2025
NXP Semiconductors Leuven, Belgium

- Build a low-cost demonstration device of an ElectroMagnetic Fault Injection (EMFI) attack against a TI LAUNCHXL-CC2640R2 board.
- Replicated a researcher's attack against the crypto core. Extended it to target a vulnerability in the bootflow sequence to re-gain JTAG communication.
- Skills: Vulnerability analyst, Python, Chipwhisperer, Side-Channel, 3D printing, Soldering, EMFI, EM, Embedded Systems, Reverse Engineering, JTAG, Fusion 360, KiCad

INDEPENDENT IT CONSULTANT (Junior Enterprise) April 2024 - July 2025
Junior Consulting Louvain Louvain-la-Neuve, Belgium

- Digital marketing consultant advising small local business.
- Skills: SEO, Odoo suite, Digital Marketing, Google Analytics, Wordpress, ERP

PROJECTS

Master thesis - SANDA: Self-Adaptive Variable-Precision Accelerator for Efficient LLM Inference June 2025 - June 2026

- Working on state of the art Hardware-aware quantization with Hardware/Software co-design for ASICs.
- Online FP mantissa-length search achieving $2.37\times$ throughput increase incurring sub-1% PPL degradation on SoTA LLMs.
- End-to-end LLM inference on-chip using KV-cache reducing memory usage by $2.46\times$.
- Received the grade of 18/20 for this Master thesis
- Skills: Python, PyTorch, SystemVerilog, Quartus, Genus, KV-cache, Research

Senior year Electrical Engineering Projects September 2025 - June 2026

- Design of a class-A Power Amplifier with $P_{sat} = 22.51dBm$ and ripple in linear region below $0.08dB$ and $PAE_{max} = 43.66\%$
- Build an TD3 based optimizer for a two-stage OTA reaching pareto-optimal results. Embedded Generative AI for explanation of the design and refinement.
- GeMM accelerator using a $4 \times 4 \times 4$ MACs systolic array reaching high spatial utilization and arithmetic intensity.
- Skills: RF-mm wave Power Amplifier design, Analog design, Cadence, Python, Generative AI, Optimization, Verilog, Test and Verification, Direct mapped memory, Quartus

EXTRACURRICULAR

AI-HDL Competition: Selected participant of the AI HDL competition hosted by the University of Arizona. 2026

CyberSecurity Challenge: semi-finalist and finalist of the 2024, 2025, and 2026 edition of the CyberSecurity Challenge. A CTF focused on Cybersecurity hosted at the Royal Military Academy. 2024-2025

UCLouvain Down: built a website that monitors the status of UCLouvain's services. Improved my knowledge in web development, digital strategy, database and network. 2024